

A3 Coursework Report

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1 A2 Weight sharing layers

Word Count: 499

1.1 Introduction

This section proposes a custom convolutional layer implemented in `Keras`, explicitly enforcing equivariance under discrete rotations $\{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$. Performance is benchmarked against a standard multilayer perceptron (MLP) on the MNIST dataset [1]. Two experiments are conducted:

1. **Experiment 1:** Assessing model generalization to unseen rotations.
2. **Experiment 2:** Evaluating performance when rotations are included during training.

Representative MNIST images subjected to discrete rotations are illustrated in Figure 1

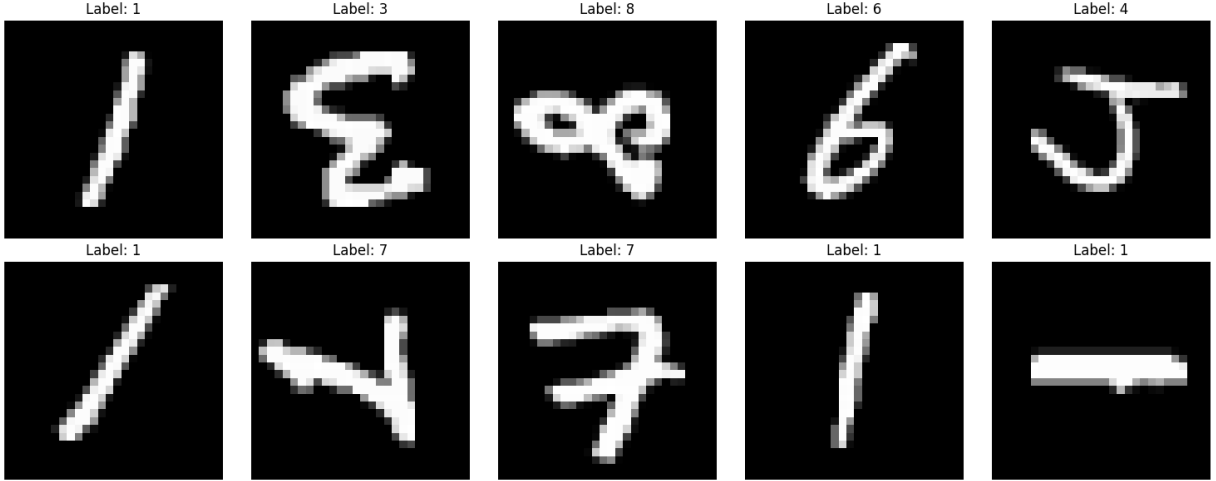


Figure 1: Example MNIST images with and without rotations.

1.2 Rotational Equivariance Layer

The rotation-equivariant convolutional layer (`RotationEquivariantConv`) achieves equivariance under the discrete cyclic rotation group $G = \{r_0, r_1, r_2, r_3\}$, where in input space r_m denotes rotation by $90^\circ \times (m \bmod 4)$.

The layer is constructed as follows: given an input x , feature maps are computed by applying the shared convolution operator φ to each rotated variant $r_m x$, where $r_m \in G$. These resulting feature maps are then concatenated along the channel dimension (see Fig. 2).

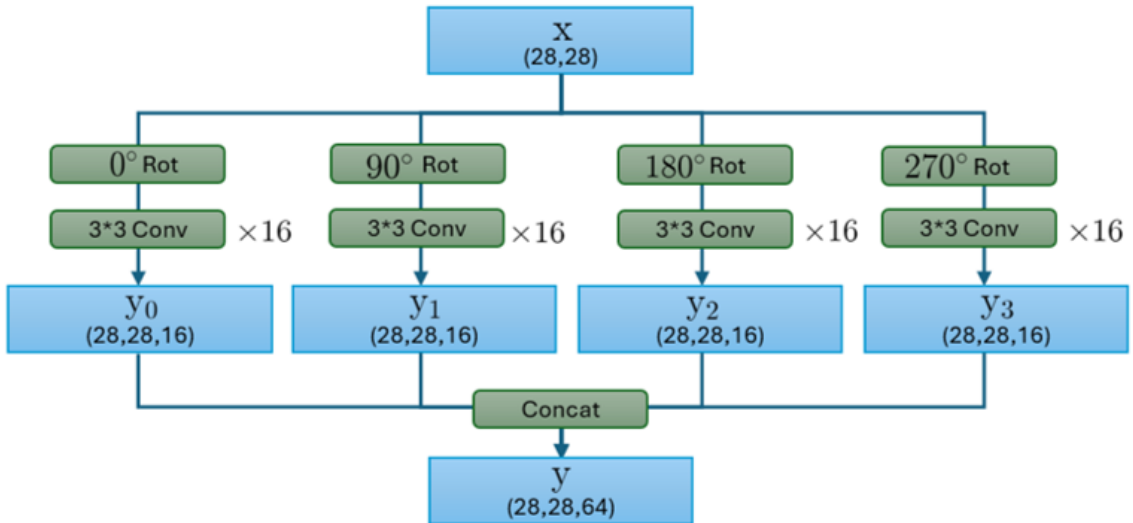


Figure 2: Equivariance convolutional layer.

1.2.1 Proof of Equivariance

Define the discrete rotation group $G = \{r_0, r_1, r_2, r_3\}$, where the group operation satisfies $r_m r_n = r_n r_m = r_{(m+n) \bmod 4}$. In input space, the action of r_m on image x is defined as rotation by $90^\circ \times m$. In output space, the action of r_m on the 4-tuple $[Y_0, Y_1, Y_2, Y_3]$ is defined as a cyclic permutation of components:

$$r_m[Y_0, Y_1, Y_2, Y_3] = [Y_{(0+m) \bmod 4}, Y_{(1+m) \bmod 4}, Y_{(2+m) \bmod 4}, Y_{(3+m) \bmod 4}].$$

We construct the map using a convolution operator φ with shared weights across orientations:

$$\Phi(x) = [\varphi(r_0 x), \varphi(r_1 x), \varphi(r_2 x), \varphi(r_3 x)].$$

To show Φ is G -equivariant, we must verify $\Phi(r_m x) = r_m \Phi(x)$:

Consider LHS:

$$\Phi(r_m x) = [\varphi(r_0(r_m x)), \varphi(r_1(r_m x)), \varphi(r_2(r_m x)), \varphi(r_3(r_m x))].$$

Using the group operation in input space $r_k r_m = r_{(k+m) \bmod 4}$, we have:

$$\Phi(r_m x) = [\varphi(r_{(0+m) \bmod 4} x), \varphi(r_{(1+m) \bmod 4} x), \varphi(r_{(2+m) \bmod 4} x), \varphi(r_{(3+m) \bmod 4} x)].$$

Consider RHS:

$$\begin{aligned} r_m \Phi(x) &= r_m [\varphi(r_0 x), \varphi(r_1 x), \varphi(r_2 x), \varphi(r_3 x)] \\ &= [\varphi(r_{(0+m) \bmod 4} x), \varphi(r_{(1+m) \bmod 4} x), \varphi(r_{(2+m) \bmod 4} x), \varphi(r_{(3+m) \bmod 4} x)], \end{aligned}$$

where the last equality holds by definition of r_m as a cyclic permutation operator in the output space.

Since LHS = RHS, we conclude Φ is equivariant with respect to the discrete rotation group G . $\square$

1.3 Model Architectures

Two distinct models utilizing different approaches are compared:

Baseline MLP: The baseline model processes flattened grayscale MNIST images (28×28 pixels) with the fully-connected structure illustrated in Figure 3:

$$\text{Input} \rightarrow \text{Flatten} \rightarrow \text{Dense}(128, \text{ReLU}) \rightarrow \text{Dense}(64, \text{ReLU}) \rightarrow \text{Dense}(10, \text{Softmax}).$$

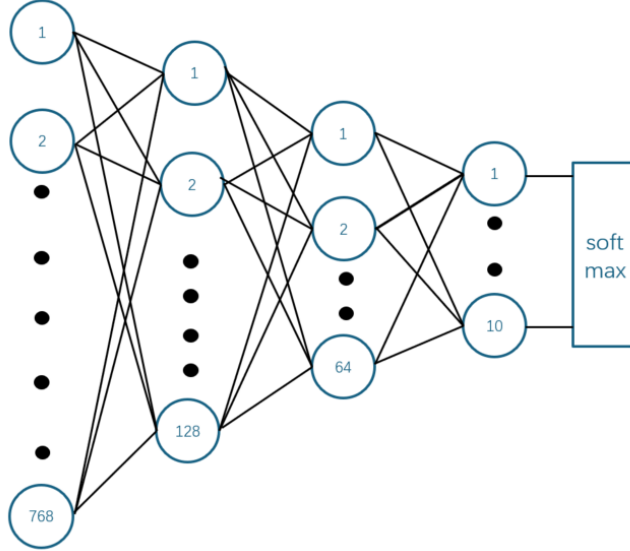


Figure 3: Baseline MLP architecture.

Rotation-Invariant CNN: Building on the rotation-equivariant layer described in Figure 2, rotation invariance is obtained by orientation average pooling. After applying 16 shared 3×3 filters to each rotated input, the resulting $(B, H, W, 4F)$ tensor is reshaped to $(B, H, W, 4, F)$ and averaged along the orientation axis. The full architecture therefore

$$\text{Input} \rightarrow \text{RotEquivConv} \rightarrow \text{OrientationAvgPool} \rightarrow \text{Flatten} \rightarrow \text{Dense}(64, \text{ReLU}) \rightarrow \text{Dense}(10, \text{Softmax}).$$

1.4 Experimental Setup

Both experiments used MNIST (50,000 training, 10,000 validation/test), trained models for 10 epochs using Adam (learning rate 0.001, batch size 128).

In Experiment 1, the training and validation sets contain only non-rotated images, while the test set includes rotated images. In Experiment 2, all datasets (training, validation, and test) include rotated images.

In Experiment 1 (Table 1), the baseline MLP achieves high training accuracy (99.0%) but severely underperforms on rotated test data (37.1%), indicating limited generalization. This discrepancy suggests the MLP memorizes orientation-specific features, unlike the Rotation-Equivariant CNN, which structurally encodes symmetry and maintains robust accuracy (94.1%) on unseen rotations.

In Experiment 2, both models achieve comparable accuracy ($\sim 93\%$), indicating that extensive rotation augmentation allows the baseline MLP to approximate invariance, diminishing the CNN’s explicit structural advantage.

Model	Experiment 1		Experiment 2	
	Train Acc	Test Acc	Train Acc	Test Acc
Baseline MLP	99.0%	37.1%	94.1%	93.5%
Equivariance CNN	96.0%	94.1%	95.5%	93.4%

Table 1: Training and test accuracy for Baseline MLP and Rotation-Equivariant CNN under two experimental setups.

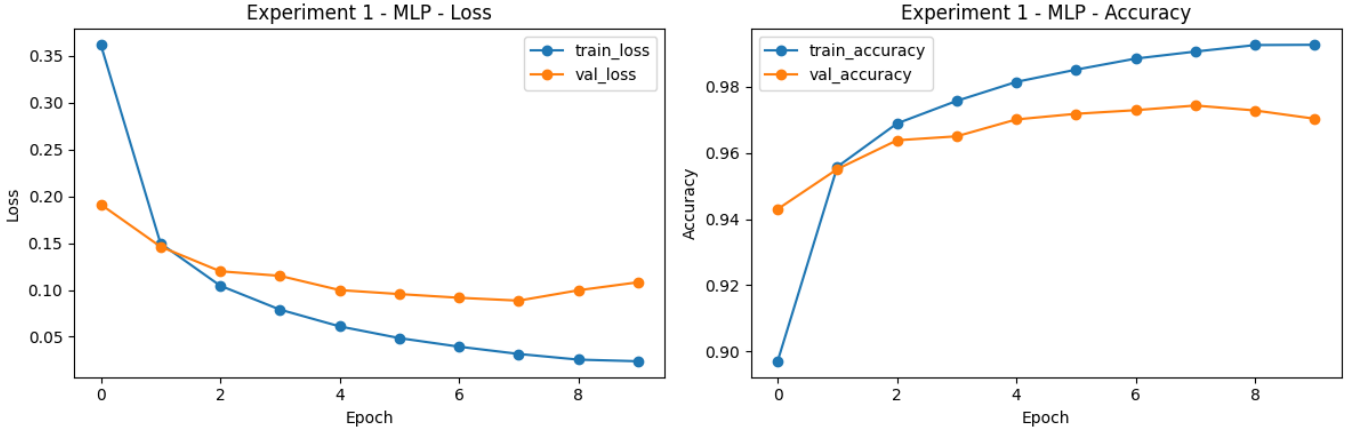


Figure 4: Baseline MLP.

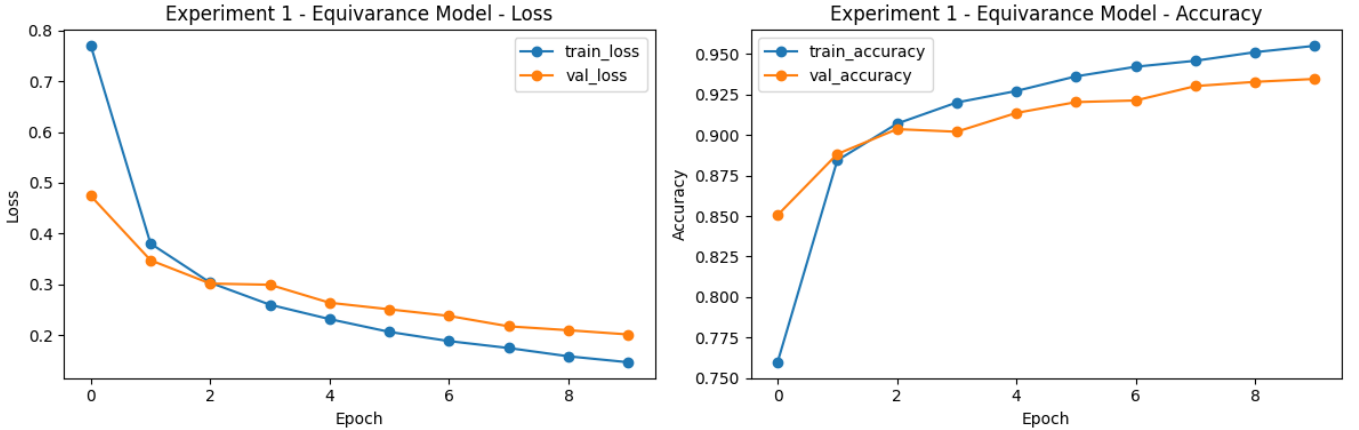


Figure 5: Rotation-equivariant CNN.

Training and validation loss/accuracy for Experiment 1.

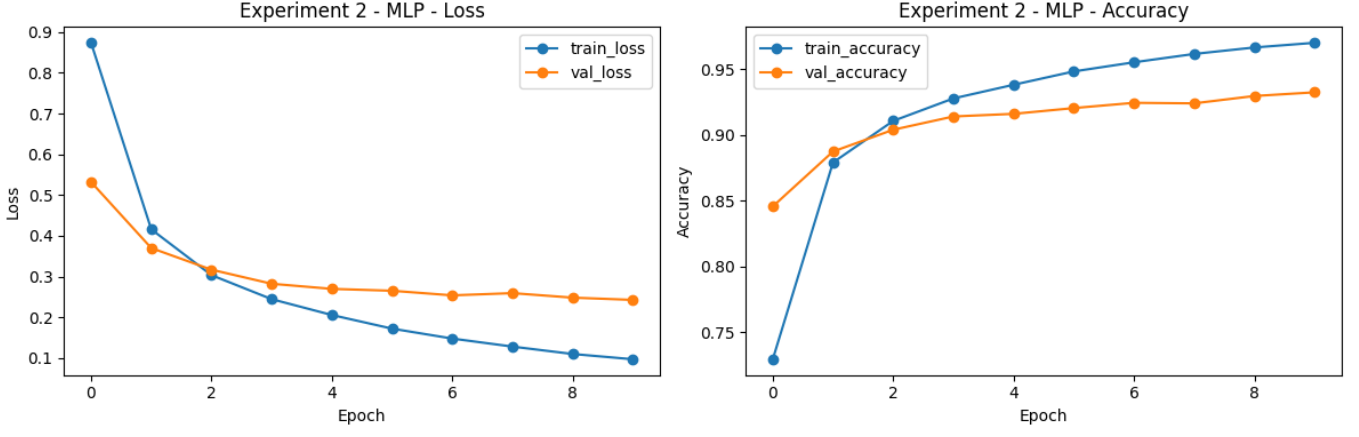


Figure 6: Baseline MLP.

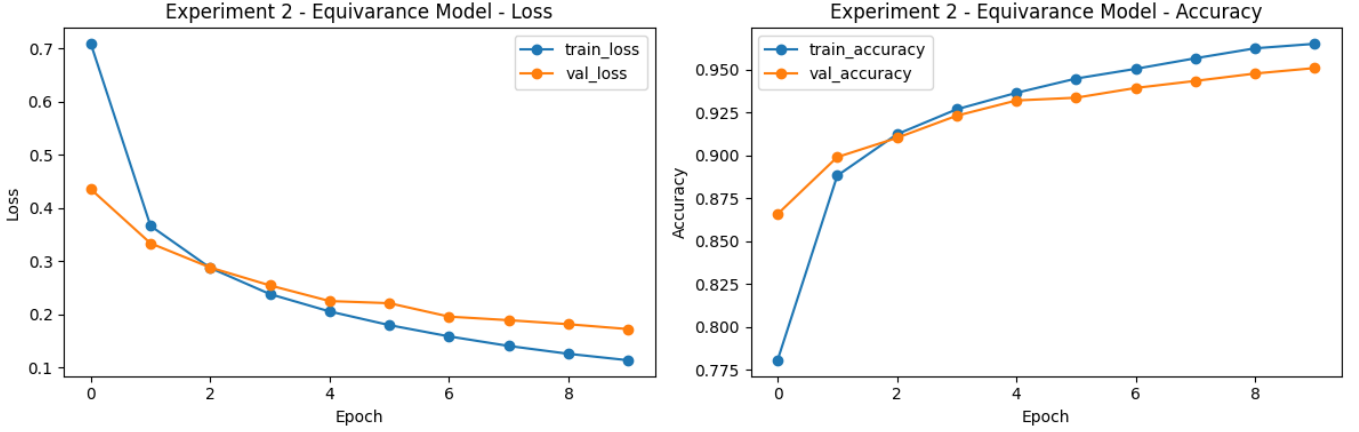


Figure 7: Rotation-equivariant CNN.

Training and validation loss/accuracy for Experiment 2.

1.5 Conclusion

Rotation-equivariant convolutional layers, combined with orientation pooling to achieve rotation invariance, significantly enhance generalization to unseen rotations, particularly when training data lacks rotation diversity. By explicitly encoding rotational symmetry, the CNN architecture attains greater robustness than the traditional MLP model, as demonstrated in Experiment 1. However, this explicit structural advantage diminishes when extensive rotation augmentation is applied during training (Experiment 2).

2 B2 More on Higgs Discovery

Word Count: 495

2.1 Introduction

This analysis uses ATLAS Open Data [2] corresponding to an integrated luminosity of 36.1 fb^{-1} . The objective is to assess the statistical significance of observing the Higgs boson decay into two photons ($H \rightarrow \gamma\gamma$).

2.2 Data Description

Two datasets (real and Monte Carlo simulation) are analyzed, structured by event and photon indices.

Real Data (36.1 fb^{-1} , $\sim 5\text{M}$ events):

- `photon_n`: photon count per event
- `photon_pt` (GeV), `photon_eta`, `photon_phi`, `photon_e` (GeV)
- `photon_isTightID`, `photon_isTightIso`: photon identification/isolation flags

Monte Carlo (10 fb^{-1} , $\sim 2.47\text{M}$ events):

- `photon_pt` (GeV), `photon_eta`, `photon_phi`, `photon_E` (GeV)
- `photon_ptcone30`, `photon_etcone20` (GeV): isolation variables
- `mcWeight`, `XSection` (pb), `SumWeights`: event normalization factors

Photon-pair invariant mass ($m_{\gamma\gamma}$) is calculated for event selection.

2.3 Event Selection

The event selection criteria are as follows:

1. Photons must pass `photon_isTightID` and `photon_isTightIso`.
2. Events must have at least two photons remaining.
3. Select the two highest- p_T photons per event.
4. Require leading photon $p_T > 50 \text{ GeV}$ and sub-leading photon $p_T > 35 \text{ GeV}$.
5. Each photon must have $p_T/m_{\gamma\gamma} > 0.35$.
6. Invariant mass selection: $100 \leq m_{\gamma\gamma} \leq 160 \text{ GeV}$.

The final selection yields 9702 events. Figure 8 shows the resulting diphoton invariant mass distribution.

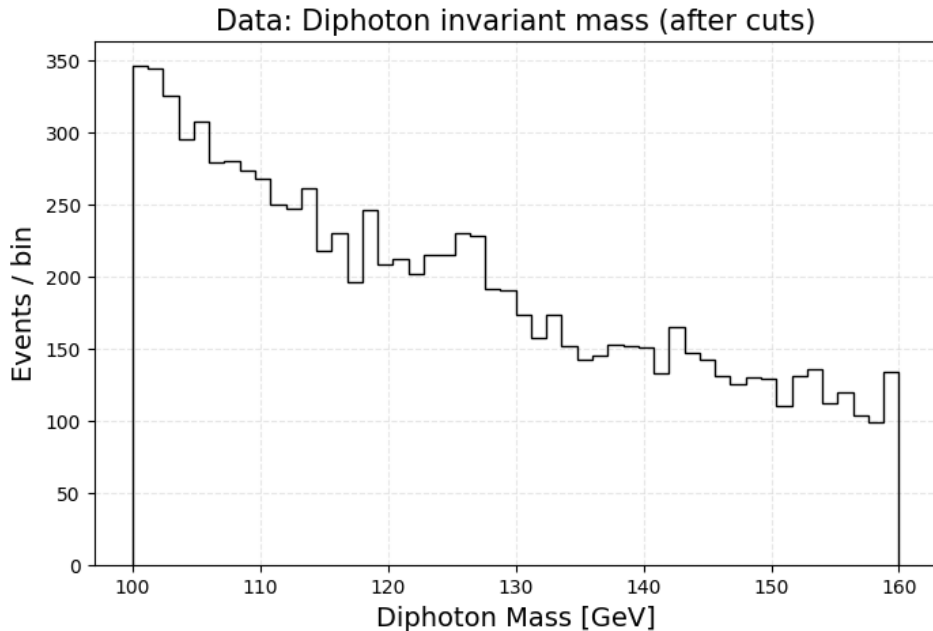


Figure 8: Diphoton invariant mass distribution after selection.

While ideally identical selections apply to experimental and MC data, for computational simplicity, MC events with photon count not equal to two are discarded, affecting only 1.6% of events. All other selection criteria—photon identification, transverse momentum thresholds, $p_T/m_{\gamma\gamma}$ ratio, and invariant mass range—match those applied to experimental data, except for photon isolation variables due to dataset constraints.

MC event weights are normalized to approximate detector-level event yields under stricter isolation criteria at 36.1 fb^{-1} . This normalization provides the Standard Model predicted signal event count (N_{sig}) and does not affect the statistical significance.

Figure 9 shows the weighted MC Higgs mass distributions used for fitting the signal model.

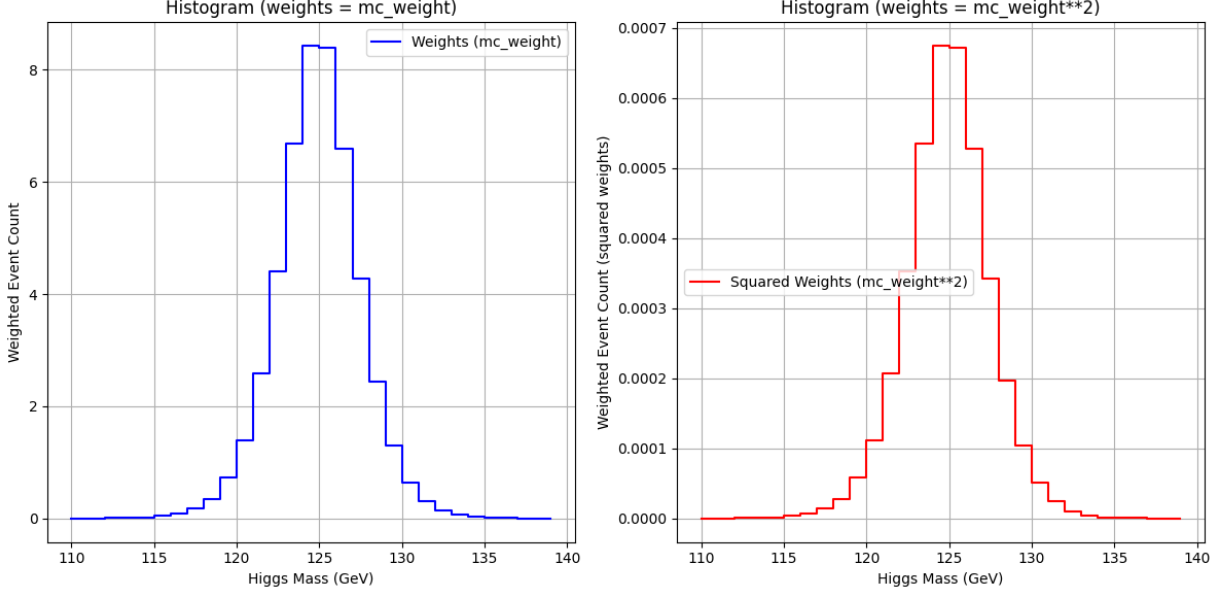


Figure 9: Weighted MC Higgs mass histograms: event weights (left) and squared weights (right).

2.4 Signal and Background Fitting Procedure

Signal Model: The signal probability density function (PDF) is modeled as Gaussian plus Crystal Ball function:

$$f_{\text{sig}}(x; \mu, \sigma_1, \sigma_2, f, b, m) = f \frac{1}{\sqrt{2\pi} \sigma_1} \exp\left[-\frac{(x-\mu)^2}{2\sigma_1^2}\right] + (1-f) \text{CB}(x; \mu, \sigma_2, b, m), \quad (1)$$

with peak μ , widths σ_1, σ_2 , tail parameters (b, m) , and Gaussian fraction f .

Background Model: The background PDF is described by a 4th-order Bernstein polynomial over the mass range $[100, 160]$:

$$f_{\text{bkg}}(x; \{b_i\}) = \sum_{i=0}^4 b_i B_{i,4}(x), \quad (2)$$

with parameters b_i from the fit.

Total Model: Signal-plus-background PDF:

$$f_{\text{tot}}(x) = N_{\text{bkg}} f_{\text{bkg}}(x) + \mu_{\text{sig}} N_{\text{sig}} f_{\text{sig}}(x), \quad (3)$$

with background normalization N_{bkg} , signal expectation N_{sig} , and strength μ_{sig} .

Hypothesis Testing: Two hypotheses considered:

- H_0 (Background-only): $\mu_{\text{sig}} = 0$.
- H_1 (Signal+Background): free μ_{sig} .

The predicted number of events per bin, ν_i , according to the total model, is compared to the observed counts, n_i , to construct an extended binned negative log-likelihood (NLL):

$$-2 \ln \mathcal{L} = 2 \sum_i \left[\nu_i - n_i + n_i \ln \left(\frac{n_i}{\nu_i} \right) \right]. \quad (4)$$

Fitting Steps: The fitting procedure minimizes the negative log-likelihood (Eq. 4) and following fits are performed:

1. Fit signal model (Eq. 1) to MC data to obtain signal shape parameters $(\mu, \sigma_1, \sigma_2, b, m, f)$.
2. Fit background-only model (Eq. 2) to data with $\mu_{\text{sig}} = 0$, obtaining $\ln \mathcal{L}_{\text{bkg}}$.
3. Fit total model (Eq. 3) to data with free μ_{sig} , obtaining $\ln \mathcal{L}_{\text{sig+bkg}}$.

2.5 Fit Results and Significance

The log-likelihood ratio test statistic is:

$$-2 \ln \frac{\mathcal{L}_{\text{bkg}}}{\mathcal{L}_{\text{sig}+\text{bkg}}} = 13.73, \quad (5)$$

corresponding to a significance of 3.71σ (Wilks' theorem [3]), strongly indicating the Higgs signal.

Figure 10 shows the likelihood scan for μ_{sig} , with a best-fit value of 1.06, slightly above the Standard Model expectation ($\mu_{\text{sig}} = 1$), suggesting minor discrepancies between MC simulation and data.

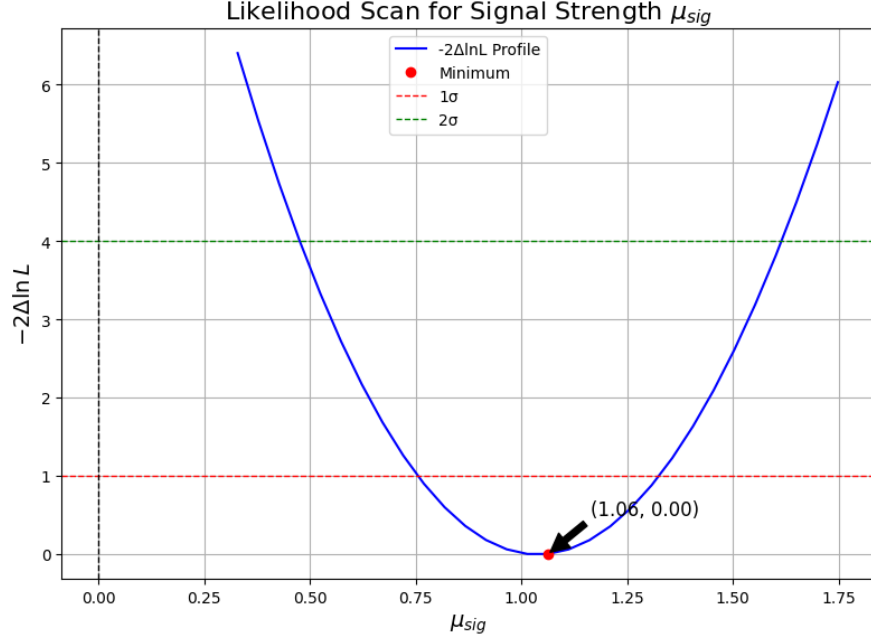


Figure 10: Profile likelihood scan for the signal strength parameter μ_{sig} . Horizontal lines represent 1σ (red) and 2σ (green) confidence intervals.

Figure 11 shows a clear excess of events at approximately 125 GeV above the background prediction, indicating a Higgs signal. The residuals confirm that the excess closely matches the expected signal shape obtained from the MC simulation.

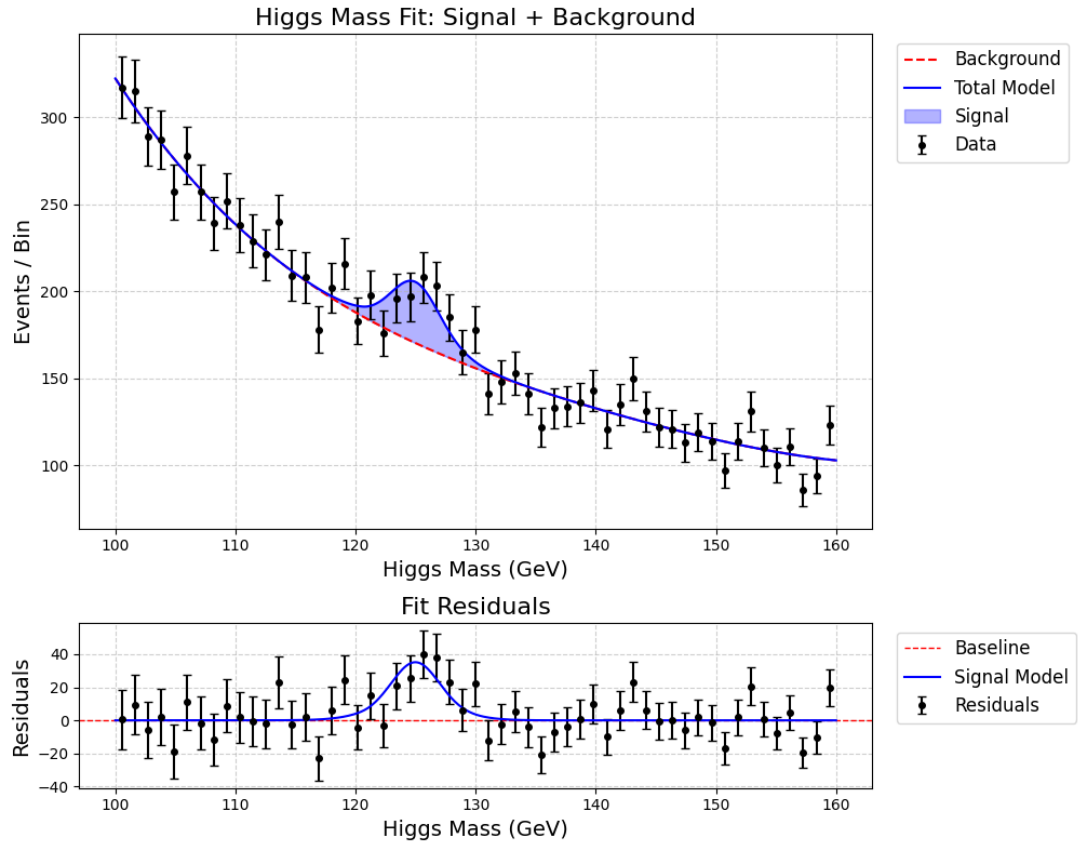


Figure 11: Upper panel: Diphoton invariant mass distribution (black points), background-only fit (red dashed), total signal-plus-background fit (blue line), and the fitted Higgs signal contribution (blue shaded area). Lower panel: Residuals between data and background model, highlighting the signal shape near 125 GeV (blue line).

3 C2 Event Classification with Neural Networks

Word Count: 1982

3.1 Introduction

Jet-flavor tagging—identifying the originating quark flavor from particle jets—is crucial for precision physics analyses. This study focuses on classifying hadronic decays of the Z^0 boson produced at a center-of-mass energy (≈ 91.2 GeV) in simulated electron-positron collisions at the Future Circular Collider (FCC) [4]. Specifically, a Transformer-based neural network is developed using to discriminate among $Z^0 \rightarrow b\bar{b}$, $c\bar{c}$, and $s\bar{s}$ decays using only measurable final-state particle and vertex-level features. The performance and generalization capabilities of the model are evaluated, emphasizing physically measurable features.

3.2 Physics Background and Motivation

High-energy e^+e^- collisions near ≈ 91.2 GeV (the Z -pole) produce Z^0 bosons that decay into quark-antiquark pairs. These quarks hadronize rapidly into collimated sprays of particles (jets). Different quark flavors leave distinct signatures:

- **b -jets:** The long lifetime of B hadrons results in secondary vertices that are significantly displaced from the primary collision point.
- **c -jets:** D hadrons from charm quarks decay faster than B hadrons, so their secondary vertices appear closer to the primary vertex.
- **s -jets:** Jets from strange quarks produce prompt, lighter hadrons without clear secondary vertices.

This analysis distinguishes the decay channels:

$$Z^0 \rightarrow b\bar{b}, \quad Z^0 \rightarrow c\bar{c}, \quad Z^0 \rightarrow s\bar{s}$$

using only final-state particle information (proton, antiproton, K^- , π^- , μ^+ , positron, electron, μ^- , photon, K_L^0 , π^+ , K^+ , neutron) and vertex data obtained from simulations.

3.2.1 Detector

A typical modern detector (Figure 12) consists of four primary subsystems:

- **Tracking System:**
Measures trajectories of charged particles (electrons, positrons, muons, charged hadrons) to determine their momentum, charge, and vertex positions via curvature in a magnetic field.
- **Electromagnetic Calorimeter (ECAL):**
Measures energy deposits of electrons, positrons, and photons through electromagnetic showers to determine their energies.
- **Hadronic Calorimeter (HCAL):**
Measures energy deposits from hadrons (protons, neutrons, antiprotons, antineutrons, K_L^0 , $\pi^\pm$, $K^\pm$) through hadronic interactions to determine their energies.
- **Muon Spectrometer:**
Identifies muons ($\mu^\pm$), measuring their momentum and confirming their identity through their penetrating behavior.

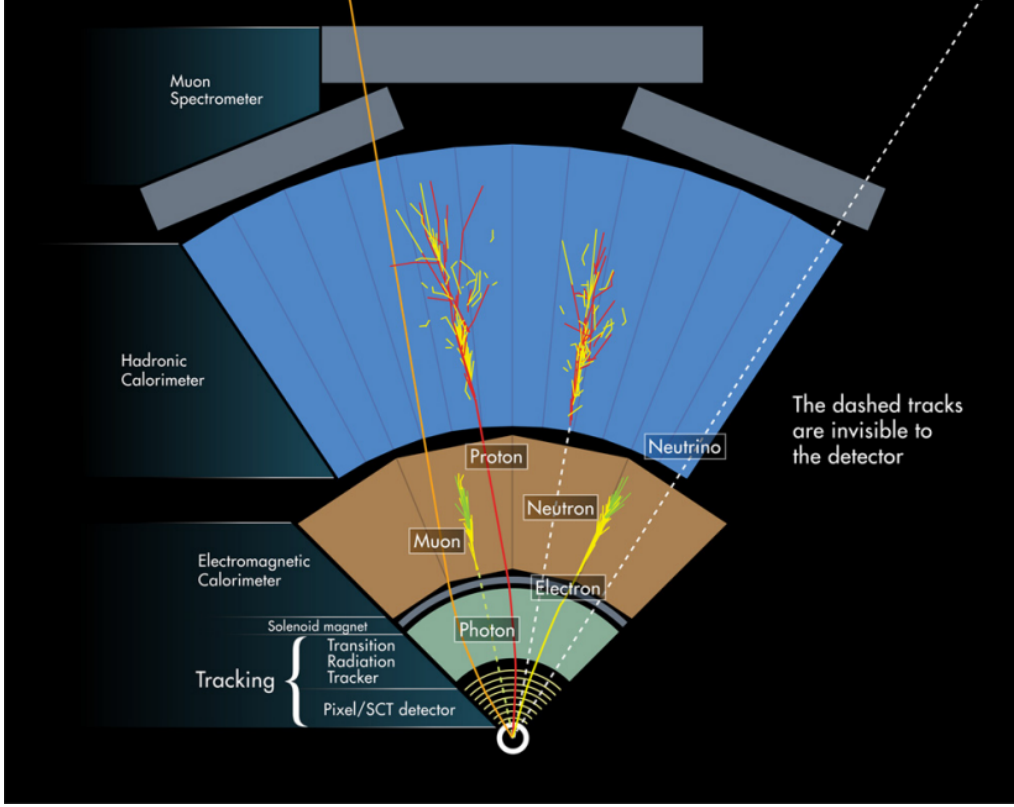


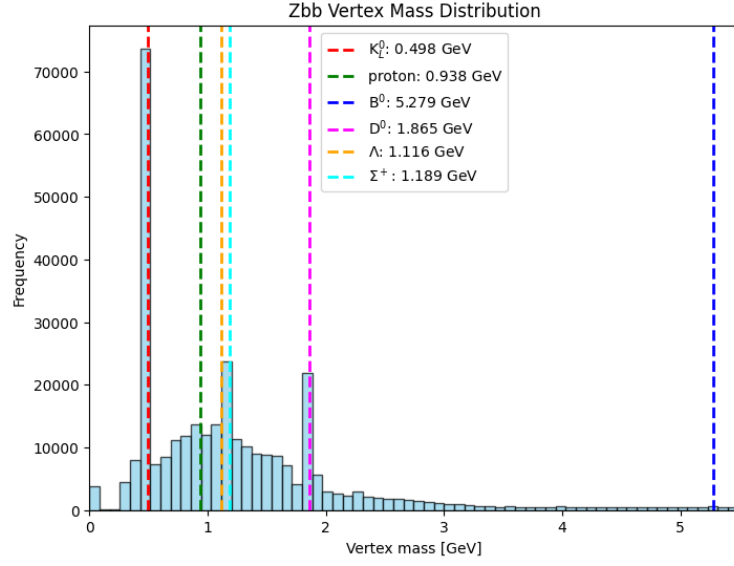
Figure 12: Schematic cross-section of a typical particle detector. Particles traverse successive detector layers, leaving characteristic signatures for identification. [5]

3.3 Data and Preprocessing

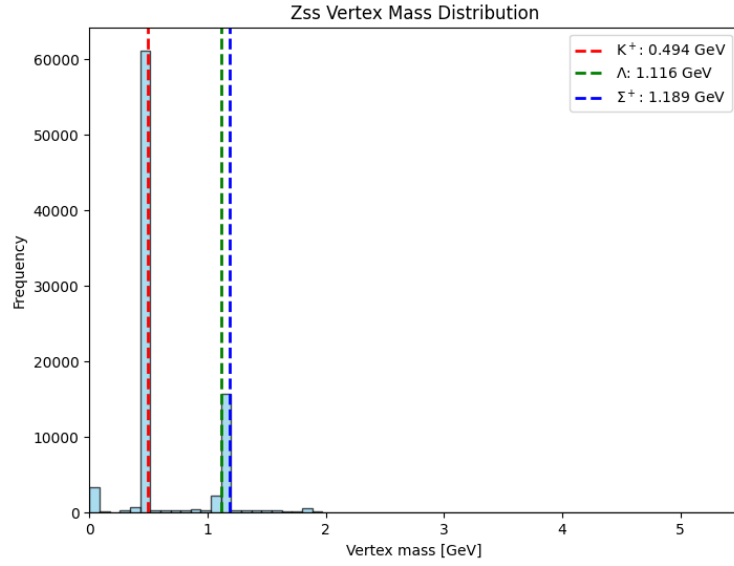
Simulated datasets Zbb , Zcc , and Zss each contain approximately 200,000 events, including particle, vertex, and thrust information structured as DataFrames indexed by `entry` (stands for events) and `subentry`. For computational efficiency, subsets from each dataset were selected: 50,000 events per flavor for training (150,000 total), and 15,000 events per flavor for validation and testing (45,000 total each). Events consist of reconstructed final-state particles, vertices, and a global thrust axis. Cartesian components of this thrust axis represent the primary direction of particle jets, closely aligned with the original quark-antiquark axis.

The original dataset records only final-state particles, omitting intermediate hadrons produced directly by the hadronization of $b\bar{b}$, $c\bar{c}$, or $s\bar{s}$ pairs. Nevertheless, typical hadronization products can be indirectly inferred from the vertex mass distributions shown in Figures 13a, 13c, and 13b:

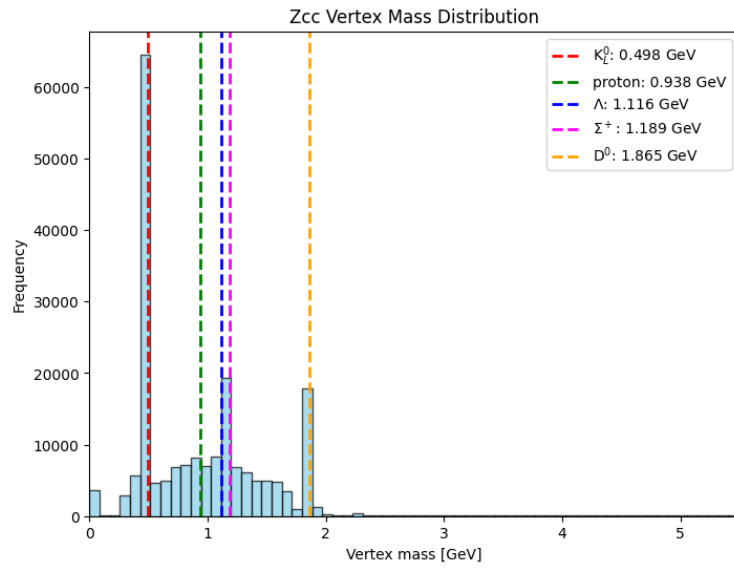
- $b\bar{b}$ events: mesons (B , D , K) and baryons (proton, Λ , Σ).
- $c\bar{c}$ events: mesons (D , K) and baryons (proton, Λ , Σ).
- $s\bar{s}$ events: mesons (K) and baryons (Λ , Σ).



(a) Zbb



(b) Zss



(c) Zcc

Figure 13: Vertex mass distributions for each $q\bar{q}$ decay channel.

3.3.1 Feature Selection

The selected particle-level features include the particle count per event (n_{particle}), particle mass m (GeV/c^2), charge q ($-1, 0, 1$), energy E (GeV), total momentum magnitude p (GeV/c), transverse momentum p_T (GeV/c), thrust error (computed from the three Cartesian thrust error components), pseudorapidity η , azimuthal angle ϕ , and the angle relative to the thrust axis. The Cartesian momentum components (px , py , pz) are excluded due to redundancy in momentum information. Particle identification (`Particle_ID`) and vertex origin indices (`Particle_orivtx_ind`) are treated separately, and their contributions are explicitly evaluated in Section 3.6.

Vertex-level features include the total number of reconstructed vertices per event, vertex coordinates (x, y, z) (mm), invariant mass (GeV/c^2), track multiplicity (the number of charged tracks originating from the vertex), vertex fit quality quantified by χ^2 , positional uncertainty (combined from individual components `Vertex_xerr`, `Vertex_yerr`, and `Vertex_zerr`) (mm), and a binary flag (`Vertex_isPV`) indicating whether the vertex is primary. Vertex fit quality (χ^2) quantifies reconstruction accuracy: heavy-flavor (b , c) decays typically produce clear secondary vertices with lower χ^2 , while light-flavor (s , u , d) vertices are less distinct, resulting in higher χ^2 . Positional uncertainties are similarly smaller for heavy-flavor vertices due to tighter track constraints.

3.3.2 Data Preprocessing

To ensure features are normalized, bounded, and efficiently learnable by the neural network, the following preprocessing steps are applied:

1. **Event-level shuffle and NaN filling:**

Events are shuffled randomly, and any remaining NaN entries are replaced by median values on a column-by-column basis.

2. **Logarithmic transformations:**

Features that are strictly non zero and span multiple orders of magnitude—such as energy E , momentum magnitude p , transverse momentum p_T , and certain error-related quantities (`Thrust_err`, `Vertex_chi2`, `Vertex_err`)—are transformed using natural logarithms:

$$x \rightarrow \log(x + \epsilon), \quad \epsilon \text{ small to avoid } \log(0).$$

3. **Angle parametrization:**

Angular variables (e.g. azimuthal angle ϕ , angle to thrust axis) are mapped to $(\sin \phi, \cos \phi)$ to mitigate discontinuities at $0/2\pi$ boundaries.

4. **Outlier clipping:**

Extreme outliers in selected features (e.g., $\log(\text{Thrust_err})$, $\log(\text{Vertex_chi2})$, and `Vertex_z`) are clipped to lie within the 0.02% to 99.98% percentile range calculated from the training dataset to remove unphysical tails.

5. **Z-score standardization:**

A selected subset of features—such as $\log(p)$, $\log(p_T)$, $\log(E)$, $\log(\text{Vertex_chi2})$, and other error-related quantities—are standardized using the mean μ and standard deviation σ calculated from the training dataset:

$$x \rightarrow \frac{x - \mu}{\sigma}.$$

Features already bounded within stable numerical ranges or angles encoded as $(\sin \phi, \cos \phi)$ are left unchanged.

6. **Fixed-length selection:**

Per event, charged particles are sorted by decreasing p_T ; up to 35 such particles are retained (supplemented by neutral particles if fewer than 35 charged ones exist). Vertices are similarly capped at 5 per event. Residual slots are zero-padded or set to `-999`.

For each event, final-state particles are sorted by descending transverse momentum (p_T) because higher- p_T objects often better reflect the dominant jet kinematics and flavor-specific decay patterns. 35 particles were retained per event—covering the average of ~ 20 charged particles while also including neutral particles to represent the total event particle multiplicity (mean ~ 43 particles per event). Charged particles generally have more accurate momentum measurements and charge assignments, aiding flavor discrimination. Similarly, vertices are limited to the top 5 per event, sorted by descending track multiplicity to prioritize vertices with clearer and more reliable reconstructions, and to surpass the average vertex count of ~ 3.4 . Remaining slots are filled with a sentinel value (`-999`) to preserve uniform input dimensions. Examples of the final distributions of selected particle-level and vertex-level features are shown in Figure 14.

After preprocessing, the shape of each event is standardized as follows:

- Particle data: (35, 14) if `Particle_ID` and `Particle_orivtx_ind` are included, otherwise (35, 12).

- Vertex data: (5, 8).

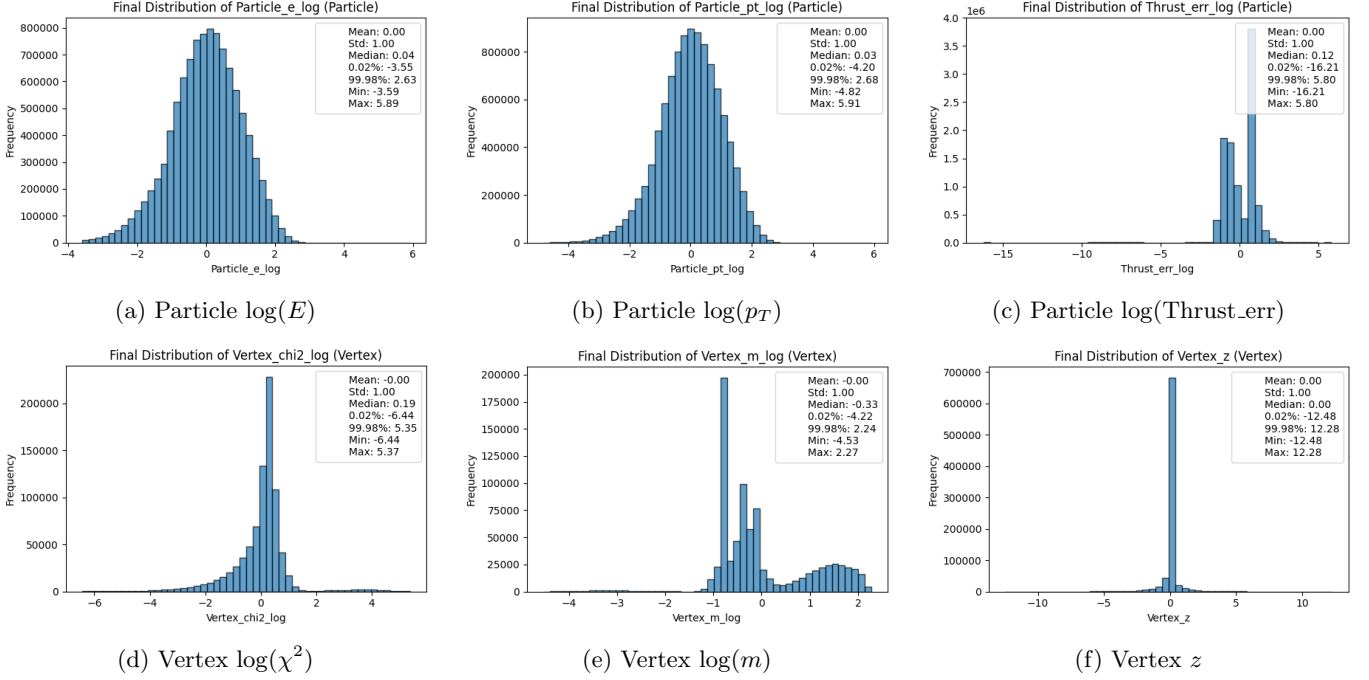


Figure 14: Final preprocessed feature distributions. The first row: particle-level features. Second row: vertex-level features.

3.4 Model Architecture

Recent studies have demonstrated strong performance of Transformer-based models in jet-tagging tasks [6]. Following this approach, the classification task here employs a Transformer architecture to process particle and vertex features in two separate input streams (Figure 15).

Both input streams undergo an initial dense embedding, projecting features into a shared dimension ($d_{\text{model}} = 64$). Particle and vertex embeddings are concatenated to form a unified sequence of length 40 (35 particles + 5 vertices).

To handle padded or invalid entries (marked by -999), a custom attention mask explicitly excludes invalid particle and vertex entries. This mask propagates through the Transformer encoder blocks and is subsequently discarded. The Transformer encoder comprises four stacked encoder layers, each containing:

1. Multi-head self-attention with 6 heads, allowing parallel exploration of subspace correlations.
2. Residual connections followed by layer normalization for improved gradient flow and training stability.
3. A position-wise feed-forward network (FFN) with an intermediate dimension of 128.

After the Transformer layers, a convolutional aggregation (Conv1D with 32 filters, kernel size 3) and global average pooling produce a fixed-length event representation, which is then fed into a dense classification head.

The final layer consists of three softmax-activated neurons, providing the predicted probabilities for each quark flavor ($Z \rightarrow b\bar{b}, c\bar{c}, s\bar{s}$).

The model is trained using categorical cross-entropy loss, with classification accuracy as the performance metric. Optimization is performed using the Adam optimizer with a learning rate of 10^{-3} .

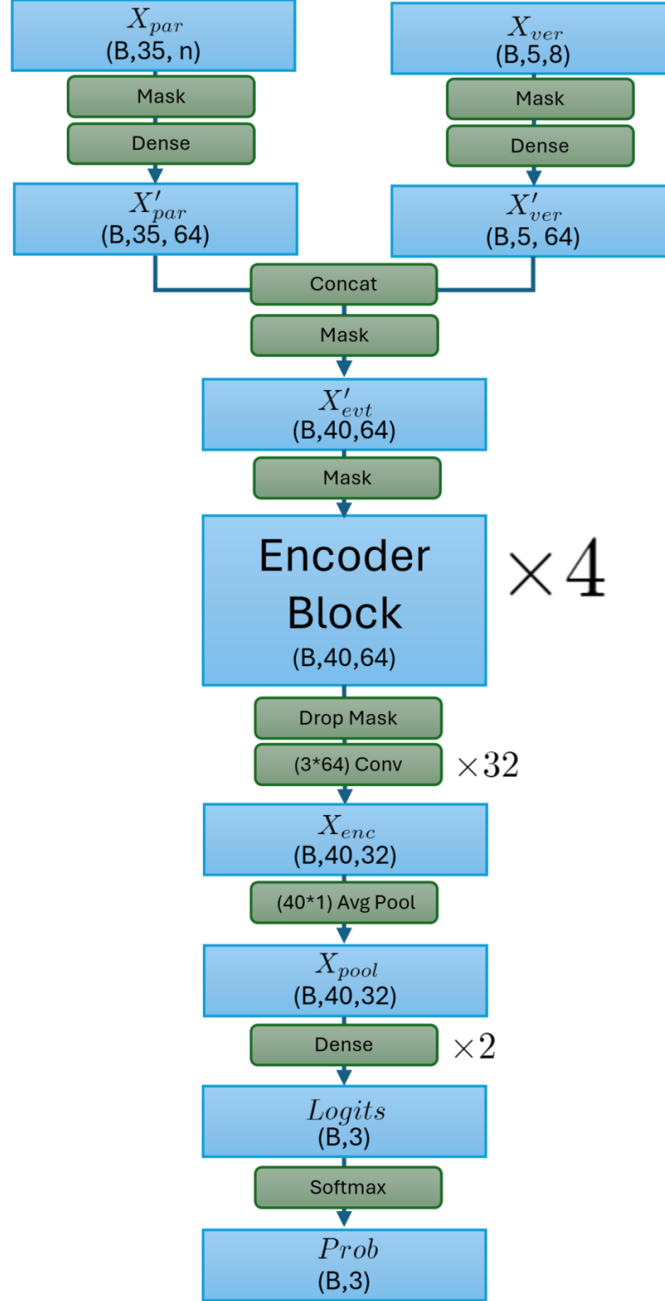


Figure 15: Schematic of the Transformer-based event classification model architecture.

3.5 Experiment and Training Setup

Two experiments are performed:

- **Exp. 1 (with ID):** Uses truth-level `Particle_ID` and `Particle_orivtx_ind` to assess leakage effects, since real experiments can only provide probabilistic estimates.
- **Exp. 2 (without ID):** Omits these two features, focusing on physically measurable quantities only.

In principle, inclusion of truth-level features such as `Particle_ID` and `Particle_orivtx_ind` can artificially enhance classification performance. Comparing these two experimental conditions quantifies the extent to which such metadata inflates results, and evaluates if physically measurable features alone yield comparable or superior performance. This distinction is essential for determining the robustness and generalizability of the flavor-tagging model under realistic experimental settings.

An initial test employed a Transformer with increased embedding dimension ($d_{\text{model}} = 128$) and six encoder layers. Despite initial progress, classification accuracy collapsed abruptly after several epochs, reverting the model to near-random performance. This instability likely stemmed from exploding gradients or unstable parameter updates. Reducing the complexity to four encoder layers, setting embedding dimension to $d_{\text{model}} = 64$, and applying gradient clipping ($\text{norm} = 1$) successfully stabilized training. Both experiments employed 50 epochs with early stopping based on validation loss.

3.6 Overall Performance

Table 2 summarizes the final classification accuracies. Surprisingly, the removal of `Particle_ID` and `Particle_orivtx_ind` (**Exp. 2**) results in higher overall performance compared to **Exp. 1**. Although these metadata may appear beneficial, their inclusion could potentially cause overfitting to simulation-specific labels, which might limit the model’s ability to generalize effectively under realistic detector conditions.

- In **Exp. 1**, early stopping was triggered at epoch 12 (Figure 16);
- In **Exp. 2**, training proceeded until epoch 36 before early stopping occurred (Figure 17).

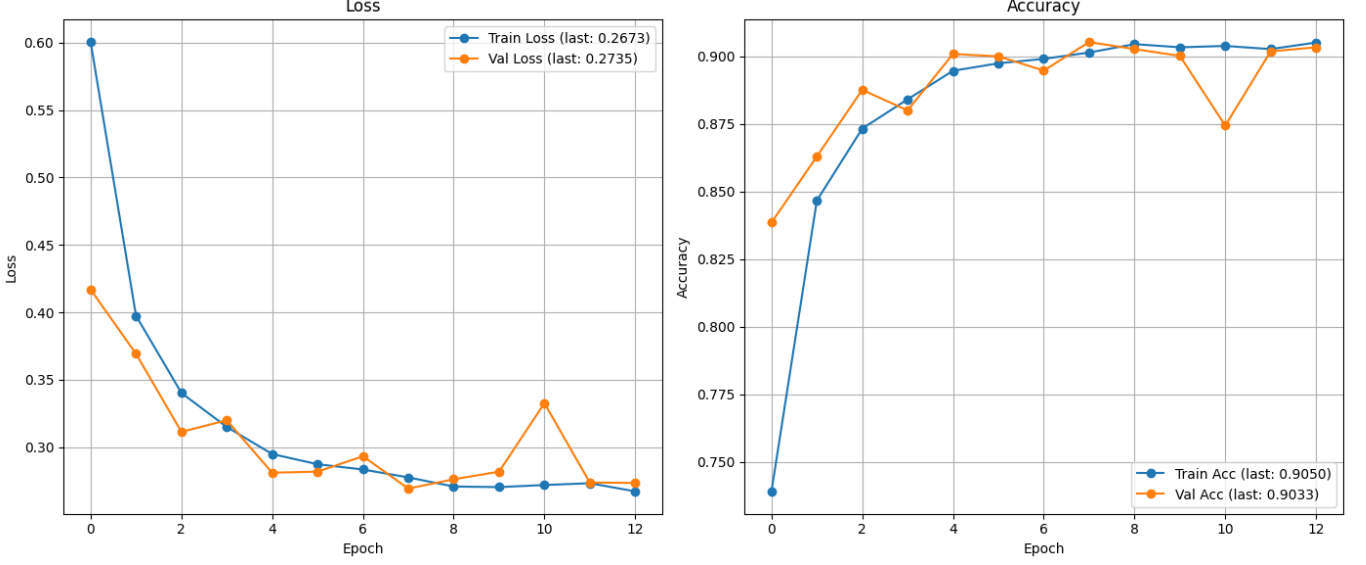


Figure 16: Loss and accuracy for Exp. 1 (with ID). Early stop at epoch 12.

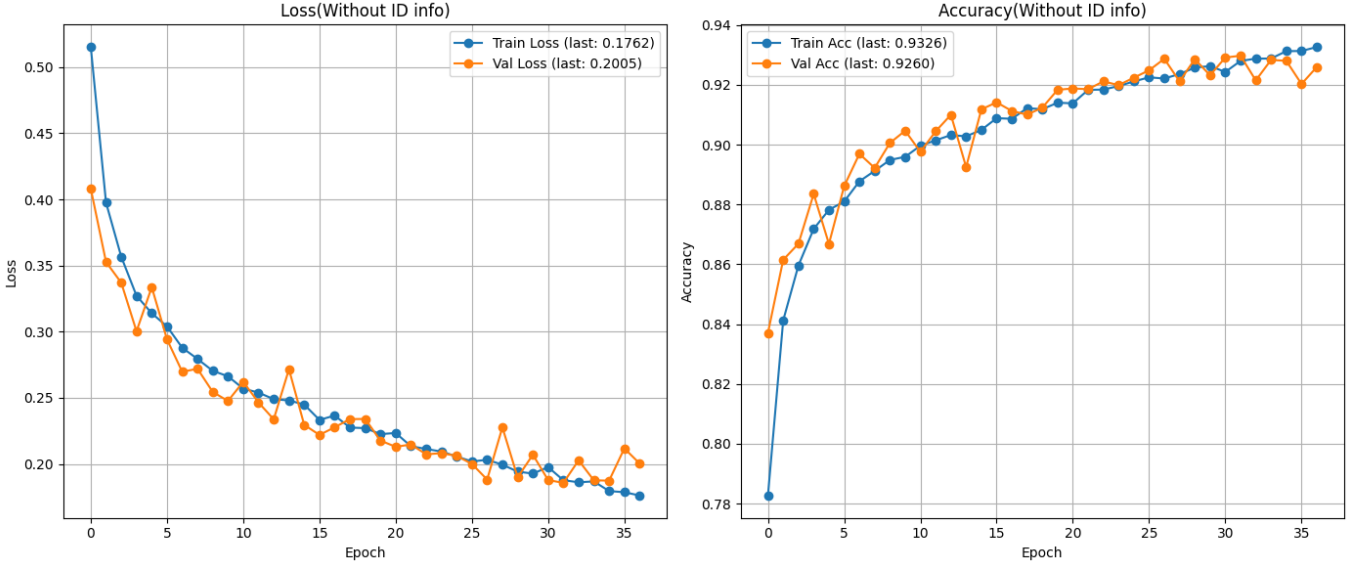


Figure 17: Loss and accuracy for Exp. 2 (without ID). Early stop at epoch 36.

Experiment	Train Acc	Val Acc	Test Acc
Exp. 1 (with ID)	0.905	0.903	0.909
Exp. 2 (no ID)	0.933	0.926	0.927

Table 2: Accuracy summary for two Transformer setups with/without ID features.

3.7 Flavor-wise Classification Performance

Distributions in Figure 18 show predicted probability densities for each flavor, obtained by applying the trained classifier separately to events of known true flavors ($Z \rightarrow b\bar{b}$, $Z \rightarrow s\bar{s}$, $Z \rightarrow c\bar{c}$). Histograms represent normalized

densities for training (bars) and test (points) datasets, with Poisson uncertainties indicated by error bars for test data.

Both experiments consistently yield the highest classification accuracy for $Z \rightarrow s\bar{s}$, followed by $Z \rightarrow b\bar{b}$, and lowest for $Z \rightarrow c\bar{c}$.

Apart from $Z \rightarrow c\bar{c}$ in Exp. 1, where the training distribution lies outside the Poisson uncertainty of the test set predictions and yields the lowest accuracy (87.1%, Figure 19) and the lowest Receiver Operating Characteristic Area Under the Curve (ROC-AUC) of 0.96, all other flavor distributions demonstrate strong consistency between training and test sets.

Distribution widths further indicate prediction certainty: narrow distributions (e.g., $Z \rightarrow s\bar{s}$) reflect clear classification signatures, whereas wider spreads (e.g., $Z \rightarrow c\bar{c}$) indicate intrinsic ambiguities in flavor identification.

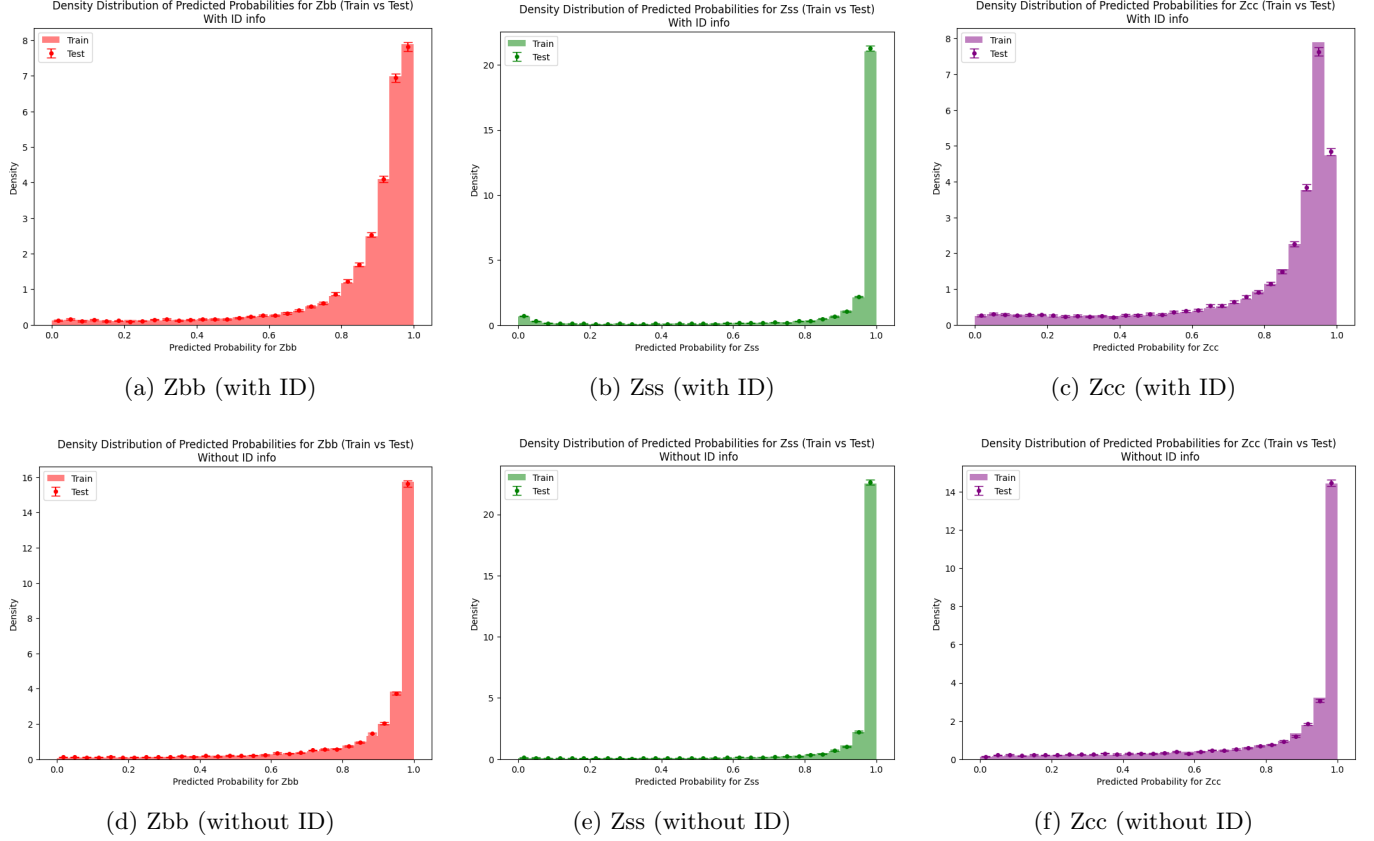


Figure 18: Density distributions of predicted probabilities for each flavor. Top row: Experiment 1 with ID features. Bottom row: Experiment 2 without ID features.

Figures 20a and 20b illustrate the ROC curves for the three-flavor classification task under the two experimental scenarios. Each curve is constructed using a one-vs-rest approach, where each flavor is treated as the positive class and the remaining flavors collectively as the negative class. Varying the classification threshold on predicted probabilities yields the True Positive Rate (TPR) and False Positive Rate (FPR), defined as:

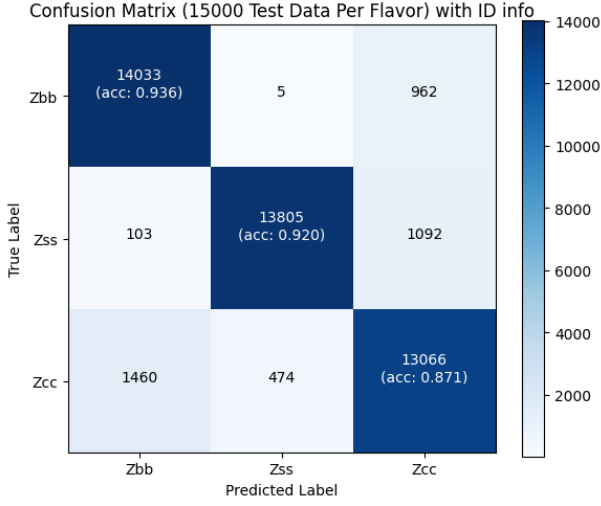
$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}, \quad (6)$$

where TP, FN, FP, and TN denote true positives, false negatives, false positives, and true negatives, respectively.

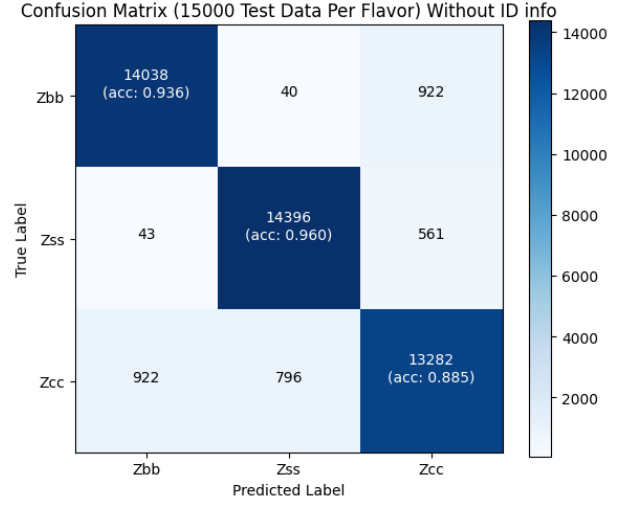
Consistent with previous accuracy analyses, the classifier achieves the highest discriminative performance for the $Z \rightarrow s\bar{s}$ channel, followed by $Z \rightarrow b\bar{b}$, and lowest for $Z \rightarrow c\bar{c}$ in both experiments.

Exp. 2 (without ID features) consistently outperforms Exp. 1 in overall test accuracy (Table 2), flavor-wise accuracy (Figure 19), and flavor-wise ROC-AUC (Figure 20). This finding underscores that relying solely on physically measurable features enhances model robustness and generalization.

Both the performance metrics and the loss/early-stopping behavior (Figures 16 and 17) reveal that including ID features causes premature saturation—training stops at 12 epochs with ID features versus 36 epochs without them.

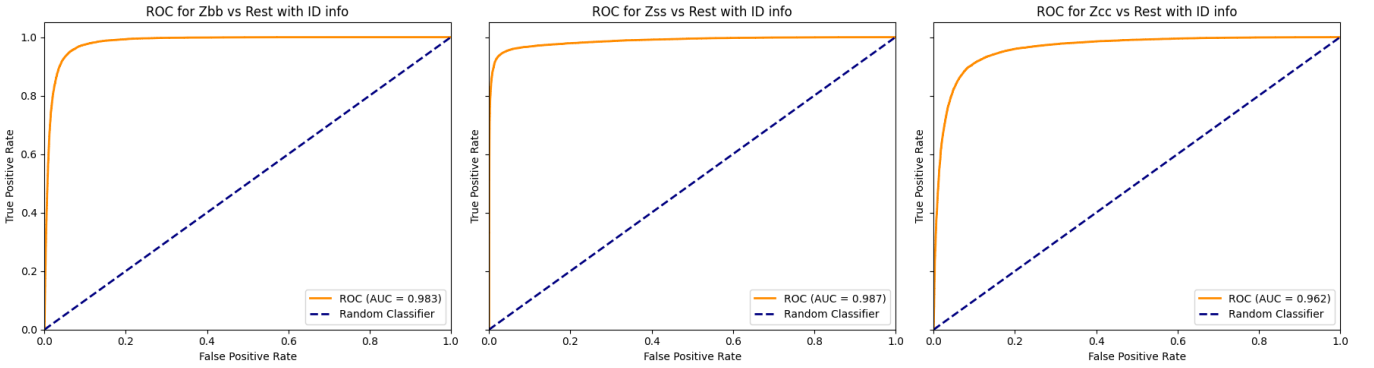


(a) Exp. 1 (with ID features)

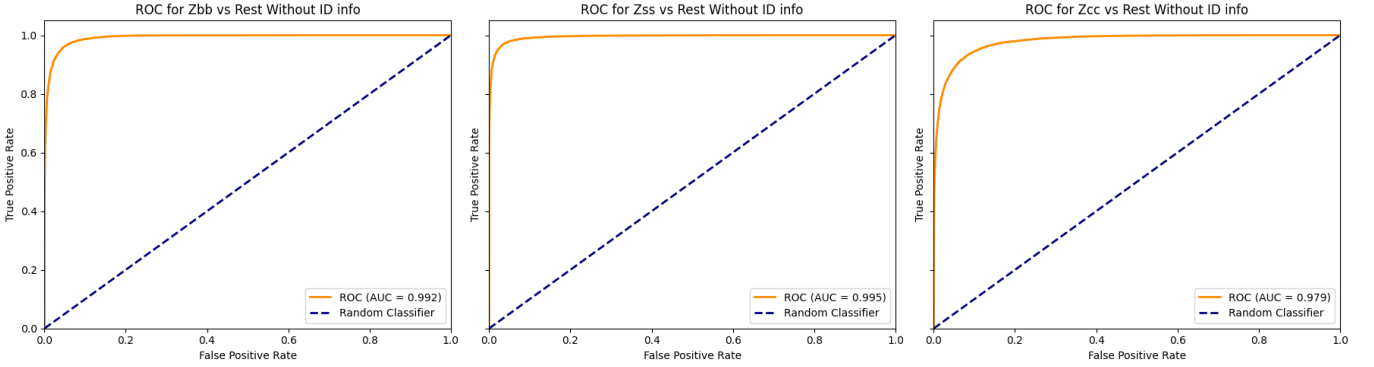


(b) Exp. 2 (without ID features)

Figure 19: Confusion matrices for Exp. 1 and Exp. 2 evaluated on test sets (15,000 events per flavor).



(a) ROC curves for Exp. 1 (with ID features).



(b) ROC curves for Exp. 2 (without ID features).

Figure 20: ROC curves for flavor classification, comparing results with (Exp. 1) and without (Exp. 2) ID-like metadata.

3.8 Discussion

Jet-flavor tagging is known to be challenging, with reported state-of-the-art accuracies of 86.1% in multi-class scenarios [6]. In contrast, three-flavor classification at the Z -pole in an idealized FCC setting achieves a test accuracy of 92.7% (Exp. 2). Several factors likely contribute to this stronger performance: (i) the Z^0 boson is nearly at rest, simplifying jet kinematics; (ii) vertex mass distributions differ notably among flavors, likely providing useful discriminative information; other features (e.g., particle multiplicity, vertex fit quality) may also be important; (iii) only three flavors (b, c, s) are considered, excluding other quark flavors and additional background processes.

Among the three channels, $Z \rightarrow s\bar{s}$ is classified most distinctly. Examination of the vertex-mass distribution for s -quark decays (Figure 13b) shows an extremely clean structure dominated by K -mesons near $0.5 \text{ GeV}/c^2$ and Λ/Σ baryons around $1.1\text{--}1.2 \text{ GeV}/c^2$, enhancing the separation of s -jets.

By contrast, $Z \rightarrow b\bar{b}$ and $Z \rightarrow c\bar{c}$ share broad and overlapping vertex-mass distributions (Figures 13a–13c), likely

causing noticeable misclassification between these two flavors (Figure 19). This overlap possibly arises because both b -quarks ($m_b \sim 4.2 \text{ GeV}/c^2$) and c -quarks ($m_c \sim 1.3 \text{ GeV}/c^2$), heavier by roughly an order of magnitude compared to s -quarks, produce sizable sets of secondary hadrons such as D -mesons, sharing similar kinematic features. Nonetheless, b -tagging slightly outperforms c -tagging, potentially due to extended heavy-meson (B) decay chains in b -jets, generating small but distinctive mass tails absent in c -jets.

3.9 Conclusion

A Transformer-based classifier was developed to distinguish Z^0 decays into $b\bar{b}$, $c\bar{c}$, and $s\bar{s}$ in an idealized FCC simulation, achieving a test accuracy of 92.7%. Notably, omitting ID features improved generalization by emphasizing physically measurable quantities.

3.10 Further Improvements

Future work should validate the importance of individual features, such as vertex mass and particle multiplicity n_{particle} , by performing an ablation study. Specifically, retraining the model without these features and comparing its performance to the baseline would clarify their contributions.

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A Appendix: Declaration of AI Usage

ChatGPT assisted with code generation from pseudocode, English grammar correction, and LaTeX formatting.